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College of Informatics and Computing Sciences

Capstone-to-ASIA Integration & Project Alignment

As previously announced, your **ASIA Semestral Project** will be based on your current Capstone Project. This is to ensure that you are not just building a standalone application, but an **Enterprise-Grade Integrated System** that follows industry standards for scalability, reliability, and interoperability.

Expected Output: A **Detailed Project Alignment Document** (PDF format) that serves as the "Architectural Blueprint" of your Capstone. This document must contain both **technical descriptions** and **visual illustrations** as specified below.

Required Content & Deliverables:

1. Project Overview & Problem Context

- Provide a concise summary of your Capstone Project.
- Clearly define the **Problem Statement** and your proposed **Technical Solution**.

2. Visual Architectural Diagram (Illustration Required)

- **Requirement:** Provide a high-level **System Architecture Diagram**.
- **Details:** This illustration must show the **3-Tier Architecture** (Presentation, Business, and Data Tiers). You must clearly label which technologies (e.g., React, Spring Boot, MySQL, etc.) will live in which tier.

3. Integration & Interoperability Plan

- **Requirement:** Aside from the text, provide a **Sequence Diagram** or **Integration Flowchart**.
- **Details:** Identify at least two (2) external integration points (e.g., Payment Gateways, Government APIs like eGovPH, SMS/Email Engines). Show how data flows from your system to these external services.

4. High Availability (HA) & Infrastructure Strategy

- **Target Uptime:** Define your goal (e.g., 99.9% vs. 99.99%).
- **Illustration Required:** A **Deployment Diagram** showing your "Cloud Jungle" setup.
- **Details:** Show where your **VPS** instances will be located. If you are aiming for HA, illustrate how your **Load Balancer** and **Redundant Servers** are positioned to prevent a single point of failure.

5. Gap Analysis (Capstone vs. ASIA)

- **Written Report:** Identify what is currently "missing" in your original Capstone Chapters 1-3 that is required for ASIA (e.g., shifting from a local-only Monolith to a cloud-ready Distributed System).